Credit Default Model — Independent Validation Report

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Executive summary

Recommendation: not approved in its current form. The model ranks credit risk acceptably (test Gini 0.490), but five High-severity findings must be resolved before it can be used for lending decisions or provisioning.

The champion model is a logistic regression that predicts whether a credit card customer will default next month. It was developed and independently validated on 30,000 customers from the UCI *Default of Credit Card Clients* dataset.

Headline results

Test	Result	Guide	Outcome
Discrimination (test Gini)	0.490	> 0.40	Pass
Separation (test KS)	0.379	> 0.30	Pass
Overfitting (train-to-test Gini drop)	0.024	< 0.05	Pass
Overall calibration (predicted vs actual)	22.0% vs 22.1%	Within 1 point	Pass
Calibration by risk band	3 of 10 bands outside ±0.03	All within ±0.03	Fail
Fairness by sex (approval ratio)	0.96	> 0.80	Pass
Fairness by age (good customers wrongly declined, 60+ vs 30–40)	22.4% vs 10.9%	Similar across groups	Fail
Response to early stress (bills +20%)	+0.1 points PD	Material increase	Fail

The five High findings

1. **The default flag may be unreliable.** 495 customers with zero bills in all six months are recorded as late, and 196 of them as defaulting.
2. **Repayment status codes are undocumented** for 58.3% of customers (codes −2 and 0).

3. **The model's largest coefficient is a suspected data artefact:** the "other" education group, which defaults at implausibly low rates.
4. **Including AGE harms older customers for no accuracy benefit.** Good customers aged 60+ are wrongly declined at twice the rate of those aged 30–40; removing AGE changes Gini by only 0.004.
5. **The model does not respond to early signs of financial stress.** A 20% rise in all bills moves average PD by just 0.1 points; only missed payments move the score.

In total the validation raised **21 findings: 5 High, 10 Medium and 6 Low**. The model's main strength is its use of a simple, explainable repayment measure (`months_late`), which proved both clearer and more accurate than the original six repayment columns.

Model overview and scope

The model estimates each customer's **probability of default (PD)** in the following month. Its intended uses are credit decisions (declining or limiting the riskiest applicants) and estimating expected losses for provisioning.

Data

Item	Detail
Source	Yeh, I. (2009), *Default of Credit Card Clients*, UCI Machine Learning Repository (doi.org/10.24432/C55S3H), CC BY 4.0
Population	30,000 credit card customers in Taiwan
Period	April to September 2005, during a regional credit card crisis
Target	`default`: 1 = defaulted the following month (6,636 customers, 22.1%)
Development / test split	70% / 30% (21,000 / 9,000), stratified by default, fixed random seed

Champion model design

- **Algorithm:** logistic regression, with standardised numeric inputs and one-hot encoded education, built as a single scikit-learn pipeline.
- **Inputs (16):** credit limit, `months_late` (number of the last six months the customer was late), six monthly bill amounts, six monthly payment amounts, age and education.

- **Excluded as inputs:** sex and marital status (protected characteristics, retained for fairness testing) and the six original repayment status columns (replaced by `months_late`).

In scope: data quality, design choices, discrimination, overfitting, benchmarking against two challengers, calibration, stability, fairness, explainability and stress testing.

Out of scope: out-of-time validation (no dates in the data), cut-off and pricing strategy, implementation and IT controls, and use of the model outside the population it was built on.

Validation approach

Each area was tested against a stated threshold, and every issue was logged with a severity in a findings register. Developer decisions were logged separately so each could be challenged.

Area	Test	Threshold used
Data quality	Missing values, duplicates, undocumented codes, implausible values, internal consistency	Any undocumented or inconsistent value is logged
Discrimination	AUC, Gini (2 × AUC − 1), KS on the test set	Gini > 0.40; KS > 0.30
Overfitting	Train vs test Gini	Drop < 0.05
Benchmarking	Challenger A (original repayment columns); Challenger B (XGBoost)	Gini gain > 0.03 treated as material
Calibration	Predicted vs actual default rate in 10 equal risk bands	Each band within ±0.03
Stability	Population Stability Index (PSI) for scores and inputs	< 0.10 stable; 0.10–0.25 investigate; > 0.25 significant
Fairness	Outcomes by sex and age at a 20% decline rate	Approval ratio ⩾ 0.80 (four-fifths benchmark); similar wrongful decline rates
Explainability	Mean absolute SHAP contribution; reason codes for individual customers	Drivers must make business sense
Stress testing	Bills +20%, payments −20%, one more month late, combined	PD should rise materially under each scenario

Severity definitions

- **High:** could materially change lending decisions, provisions or legal exposure. Must be resolved before approval.
- **Medium:** weakens accuracy, explainability or robustness. Should be resolved or formally accepted with mitigation.
- **Low:** minor, or a point to document and monitor.

The UK regulatory context considered was the PRA's model risk management principles for banks (SS1/23), the Equality Act 2010, the FCA Consumer Duty and IFRS 9 expected credit loss requirements.

Summary of results

The champion performs acceptably and is stable on held-out data, but its calibration, treatment of older customers and response to stress fall short.

Discrimination and benchmarking (test set)

Model	AUC	Gini	KS	Train Gini
Champion (`months_late`, with AGE)	0.745	**0.490**	0.379	0.514
Challenger A (six original repayment columns)	0.714	0.429	0.370	–
Challenger B (XGBoost)	0.769	0.538	0.411	0.671
Champion without AGE	0.743	0.486	0.387	–

- **`months_late` beats the original repayment columns** by 0.061 Gini, supporting the developer's design choice.
- **XGBoost is 0.048 Gini better but overfits** (train-to-test drop 0.133 vs 0.024), so it is not recommended as a replacement. Its gain shows the champion misses some patterns.
- **Removing AGE costs only 0.004 Gini.**

Calibration by risk band (test set, 900 customers per band)

Band	Predicted	Actual	Difference
1 (safest)	7.1%	5.3%	−1.7
2	11.3%	11.0%	−0.3
3	13.1%	10.7%	−2.4
4	14.3%	13.1%	−1.2
5	15.5%	13.2%	−2.3
6	16.6%	15.9%	−0.7
7	18.2%	21.0%	+2.8
8	22.9%	28.6%	**+5.7**
9	34.5%	40.9%	**+6.3**
10 (riskiest)	66.9%	61.6%	**−5.3**

Band	Predicted	Actual	Difference
All	**22.0%**	**22.1%**	+0.1

Differences are in percentage points. Overall calibration is accurate only because underestimation in bands 7–9 (about 133 more defaults than predicted) offsets overestimation in band 10 (about 48 fewer).

Stability

Score PSI between development and test samples is **0.0013**, and every input's PSI is below **0.003**. This confirms a sound split but not stability over time.

Fairness (decline the riskiest 20%; cut-off PD ⩾ 25.7%)

Group	Customers	Actual default	Predicted	Good customers wrongly declined	Approval ratio
Female	5,437	21.0%	21.6%	11.4%	1.00
Male	3,563	23.9%	22.8%	14.4%	0.96
Age 20–30	3,285	22.1%	22.7%	12.8%	0.96
Age 30–40	3,193	20.1%	20.5%	10.9%	1.00
Age 40–50	1,818	24.2%	22.4%	13.4%	0.95
Age 50–60	610	25.6%	24.8%	16.3%	0.89
Age 60–80	94	28.7%	23.4%	**22.4%**	0.89

Without AGE as an input, good customers aged 60–80 are wrongly declined at **16.4%** (down from 22.4%) and those aged 50–60 at 13.2%, while those aged 20–30 rise to 14.8%.

Stress testing

Scenario	Average PD	Change vs base	Declined at base cut-off	Score PSI vs base
Base	22.0%	–	20.0%	–
Bills +20%	22.2%	+0.1 pts	20.1%	0.001
Payments −20%	22.3%	+0.3 pts	20.2%	0.003
One more month late	29.6%	+7.6 pts	38.6%	1.154

Scenario	Average PD	Change vs base	Declined at base cut-off	Score PSI vs base
Combined downturn	30.2%	+8.1 pts	41.0%	1.292

The model responds strongly to missed payments but almost not at all to rising bills or falling payments.

High severity findings

Five findings must be resolved before approval. H1 and H2 share a root cause: the repayment status data does not behave as documented.

H1. The default flag may be unreliable (Data quality)

Finding. 795 customers have zero bills and zero payments in all six months. Of these, 495 (62%) are recorded as late in September, and 196 of those 495 (40%) are recorded as defaulting, against 22% overall.

Impact. A customer who owes nothing cannot default. Either bills are incomplete, the repayment codes mean something other than documented, or the default flag is wrong for some customers. Because the default flag is what the model learns to predict, every result in this report depends on it.

H2. Repayment status codes are undocumented for most customers (Data quality)

Finding. The September repayment status (PAY_0) takes the values −2 (2,759 customers) and 0 (14,737) for 58.3% of customers, neither of which appears in the data dictionary. Their default rates also contradict the commonly assumed meanings: customers coded −1 ("paid in full") default at 16.8%, more than those coded 0 (12.8%).

Impact. The strongest predictor in the data cannot be interpreted with confidence. The developer's `months_late` measure reduces the exposure, since it only counts codes of 1 or more, but still depends on the codes being correct.

H3. The largest coefficient is a suspected data artefact (Model design)

Finding. The "other" education group (code 4, including undocumented codes 0, 5 and 6 grouped into it) has the largest coefficient in the model (−0.93). These customers default at roughly 0–6%, against 19–25% for codes 1–3, which is not plausible as a genuine education effect.

Impact. Customers in this group receive about 60% lower odds of default based on what is likely a data capture pattern. The effect may not hold for new applicants, and an applicant could lower their score by choosing "other".

H4. Including AGE harms older customers with no accuracy benefit (Fairness)

Finding. With AGE as an input, good customers aged 60–80 are wrongly declined at 22.4%, and those aged 50–60 at 16.3%, against 10.9% for ages 30–40. Removing AGE reduces these to 16.4% and 13.2% while changing test Gini by only 0.004 (0.490 to 0.486). The highest-risk customer in the test set (PD 83.1%) did not default and had age 62 as their third-largest decline reason.

Impact. Age is a protected characteristic under the Equality Act 2010. Its use cannot be objectively justified on this evidence and creates legal and conduct risk under the FCA Consumer Duty. The 60–80 group contains only 94 test customers, but the 50–60 group (610) shows the same direction.

H5. The model does not respond to early signs of financial stress (Stress testing)

Finding. A 20% rise in all bills moves average PD by +0.1 points, and a 20% fall in payments by +0.3 points. One additional missed month moves it by +7.6 points. The bill coefficients change sign month to month and cancel each other out.

Impact. The model reacts only after customers have fallen behind, so it cannot support early intervention. For provisioning, IFRS 9 requires expected credit losses to reflect forward-looking information, which this model largely ignores.

Medium severity findings

Ten findings weaken the model's accuracy, explainability or robustness and should be resolved or formally accepted.

ID	Area	Finding	Evidence
M1	Calibration	Overall calibration hides offsetting band errors; medium-high risk customers are under-predicted	Bands 8–9 under by 5.7 and 6.3 points; band 10 over by 5.3; about 133 more defaults than predicted in bands 7–9
M2	Model design	Bill coefficients change sign month to month and cannot be explained	BILL_AMT2/3 positive, BILL_AMT1/4/5/6 negative; monthly bills correlated at 0.80–0.95

ID	Area	Finding	Evidence
M3	Explainability	A material share of each score comes from unexplainable inputs; AGE appears among decline reasons	BILL_AMT3 and BILL_AMT2 rank 3rd and 4th by mean SHAP impact
M4	Performance	The champion misses patterns a more flexible model captures	XGBoost test Gini 0.538 vs 0.490 (+0.048)
M5	Stability	Out-of-time stability cannot be tested	No dates; single six-month crisis period; PSI only confirms the random split
M6	Data quality	Undocumented education codes default at implausibly low rates	Codes 0, 4, 5 at roughly 0–6% vs 19–25% for codes 1–3
M7	Data quality	Repayment status risk order contradicts its assumed meaning	Code −1 defaults at 16.8% vs 12.8% for code 0
M8	Data quality	Some credit balances exceed the credit limit, suggesting entry errors	6 customers; one with a 280,000 limit shows −339,603 in April then reverts, and is recorded as defaulting
M9	Data quality	EDUCATION contains undocumented codes	Codes 0, 5, 6 for 345 customers (1.1%)
M10	Data quality	MARRIAGE contains an undocumented code	Code 0 for 54 customers (0.2%)

Low severity findings

Six findings are minor and should be documented and monitored.

ID	Area	Finding	Evidence
L1	Data quality	Identical rows retained as likely distinct inactive customers	35 pairs (70 rows); most show no activity in any month; no customer ID to confirm
L2	Data quality	Imbalanced target; accuracy alone would mislead	22.1% default rate; AUC, Gini and KS used instead

ID	Area	Finding	Evidence
L3	Data quality	Negative bills (credit balances) are undocumented	1,930 customers (6.4%); default rate 16.5%, below average as expected
L4	Data quality	High repayment status codes are too sparse for reliable estimates	Codes 4–8 cover 141 customers; default rates vary from 50% to 78%
L5	Fairness	Good male customers are wrongly declined slightly more often	14.4% vs 11.4%; partly reflects higher actual default (23.9% vs 21.0%); approval ratio 0.96
L6	Performance	XGBoost challenger overfits and is not recommended as a replacement	Train-to-test Gini drop 0.133 vs 0.024 for the champion

Model strengths

- **A simple, explainable main driver that also performs better.** `months_late` outperformed the six original repayment columns by 0.061 Gini and can be explained to a customer in one sentence.
- **No overfitting.** The train-to-test Gini drop is 0.024, well within the 0.05 guide.
- **Sensible core drivers.** Repayment history, credit limit and recent payments all act in the expected direction.
- **Accurate at portfolio level.** Average predicted default (22.0%) matches the actual rate (22.1%).
- **Protected characteristics handled deliberately.** Sex and marital status were excluded as inputs and kept for fairness testing.
- **Reproducible and well documented.** A fixed random seed, a single pipeline and logged developer decisions allow the results to be rebuilt exactly.

Recommendation

Not approved in its current form. The model should be returned to the developer for remediation and revalidated before any use.

The reasons are:

- **The target cannot yet be trusted (H1, H2).** A model cannot be approved when there is doubt about whether the outcome it predicts is recorded correctly. Good test metrics do not resolve this, because they are measured against the same flag.

- **Using AGE creates avoidable legal and conduct risk (H4)** for no measurable benefit.
- **The model is unsuitable for forward-looking provisioning (H5, M1)** in its current form.

Conditions for revalidation. The revised model should be resubmitted when all five High findings are resolved, with evidence that:

1. The default definition and repayment status codes are confirmed in writing by the data owner.
2. Test Gini remains above 0.40 and the train-to-test drop below 0.05.
3. Every calibration band is within ±0.03.
4. The wrongful decline rate for any age group is no more than 1.5 times that of the best-treated group.
5. A 20% rise in bills produces a material increase in average PD.

Remediation plan

Actions are listed in priority order. Owners are shown as roles, since this is a self-directed project.

#	Action	Findings addressed	Priority	Owner
1	Confirm the default definition and the meaning of every repayment status code with the data owner; correct or exclude records that cannot be confirmed	H1, H2, M7	High	Data owner and developer
2	Remove AGE as an input and rerun the fairness tests	H4, M3	High	Developer
3	Replace the six bill columns with one or two stable measures, such as credit utilisation (bill ÷ credit limit) and the trend in balance over six months	H5, M2, M3, M4	High	Developer
4	Investigate the "other" education group; group it with a documented category or remove EDUCATION if the pattern is a data capture artefact	H3, M6, M9	High	Developer and data owner

#	Action	Findings addressed	Priority	Owner
5	Recalibrate the final model by risk band (for example isotonic or Platt scaling) and confirm all bands are within ±0.03	M1	Medium	Developer
6	Rerun the stress test and confirm that rising bills and falling payments increase PD	H5	Medium	Validator
7	Correct or exclude the six customers with credit balances above their limit	M8	Medium	Data owner
8	Obtain dated data to perform out-of-time validation; until then, restrict use to the development population	M5	Medium	Model owner
9	Set up monthly monitoring using the PSI baselines in this report (alert at 0.10, escalate at 0.25) and track wrongful decline rates by sex and age	M5, L5	Low	Model owner
10	Group repayment status codes 3 and above into a single band in any future use	L4	Low	Developer

Limitations of this validation

- **Developer and validator were the same person.** This is a self-directed portfolio exercise; in practice, validation must be independent of development.
- **No access to the data owner.** Data quality findings could not be confirmed or explained at source.
- **Historical, non-UK data.** The data covers Taiwan in 2005 during a credit card crisis, so the default rate (22.1%) is far above a typical UK credit card portfolio. Results do not transfer to other populations.
- **Small subgroups.** The 60–80 age group has 94 test customers, so its fairness figures carry wide uncertainty.

- **Simplified explainability method.** SHAP values were calculated as coefficient × standardised value, which assumes independent inputs. The monthly bills are highly correlated, so their individual contributions are approximate.
- **Illustrative thresholds.** Gini, KS, calibration and PSI thresholds are common industry guides, not regulatory limits. The 20% decline rate used for fairness testing is an assumption, not a business policy.

Appendix: developer decisions and reproducibility

Developer decisions log

#	Decision	Validation outcome
1	Undocumented EDUCATION codes (0, 5, 6) grouped into 4 = "other"	Challenged: the grouped category drives the largest coefficient (H3)
2	Undocumented MARRIAGE code (0) grouped into 3 = "other"	Accepted: affects 0.2% of customers and MARRIAGE is not an input
3	Six repayment status columns replaced by `months_late`, for clarity to non-specialist readers	Supported: higher test Gini than the original columns (0.490 vs 0.429)
4	AGE retained as an input, as risk rises after 40 and lenders may use age where objectively justified	Not supported: negligible accuracy benefit and worse treatment of older customers (H4)
5	SEX and MARRIAGE excluded as inputs and retained for fairness testing	Accepted

Data source and reproducibility

- **Data:** Yeh, I. (2009). *Default of Credit Card Clients* [Dataset]. UCI Machine Learning Repository. doi.org/10.24432/C55S3H. Licensed CC BY 4.0. Loaded with the `ucimlrepo` Python package (dataset ID 350).
- **Tools:** Python with pandas, NumPy, scikit-learn, XGBoost and matplotlib.
- **Settings:** random seed 42; 70/30 stratified split.
- **Saved outputs:** trained champion model, scored test set, findings register, developer decisions, model comparison, calibration by band and stress test results.